

2023

ZERONORTH

Accelerating Sustainability in the Shipping Industry on AWS with ZeroNorth

Learn how technology provider ZeroNorth is helping shipping companies optimize operations and reduce fuel consumption on AWS.

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Scaled to serve

4,000+ vessels

5 new services

launched in 1 year

Accelerates

speed of innovation

Helps meet

sustainability targets

Supp

reduced



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Overview



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In the global fight against climate change, the shipping industry stands out as one of the most energy-intensive sectors, contributing significantly to greenhouse gas emissions. Driven by a mission to champion sustainable shipping, [ZeroNorth](#) has taken proactive measures to help shipping companies tackle these challenges. The technology provider offers a centralized solution where shipping companies can track their current data, make informed decisions, and shift toward faster, greener operations.



To bring this vision to life, ZeroNorth migrated its data and services to Amazon Web Services (AWS). By harnessing the power of AWS, the company has accelerated its speed of innovation, helping its customers drive efficiencies and meet their sustainability targets at a much faster pace.

Opportunity | Helping Shipping Companies Become Greener and More Efficient

Headquartered in Copenhagen, Denmark, ZeroNorth is a technology company with a mission to make global shipping more sustainable. Its software solution integrates services for chartering, voyaging, maintaining, and bunkering to help companies in the shipping industry reduce their operating costs and lower their greenhouse gas emissions.

The shipping industry is a major contributor to global greenhouse gas emissions, accounting for [3 percent](#) of the world's total. By 2030, emissions are [projected to rise](#) by almost 11 percent compared with levels in 2010. This statistic underscores the urgency for the shipping industry, one of the most energy-intensive sectors, to prioritize sustainable practices more than ever before.

With over 20 different legacy systems and communication methods used throughout the shipping industry, finding ways to improve efficiency and sustainability can be difficult. Because these systems are so complex and are not integrated, shipping companies often grapple with siloed information and miss opportunities for optimization.

Before ZeroNorth migrated to AWS, it hosted its solution on a different cloud provider—but soon, the need for a technology upgrade became evident. The company urgently needed to unite siloed data, connect disparate sources, and automate manual processes. After a thorough evaluation, ZeroNorth saw an opportunity to modernize some of its software components on AWS.

With agile cloud solutions from AWS, ZeroNorth could create a unified environment that could merge legacy systems and speed up the deployment of digital services. Moreover, ZeroNorth found solutions to several internal challenges on AWS, including issues related to single-tenant architecture and a previously existing gap in control policies, data management procedures, and orchestration workflows.

Solution | Expanding the ZeroNorth Solution to Support Over 4,000 Vessels and Offer Six Services

Data is at the core of ZeroNorth's mission; every data point can help drive critical sustainability decisions and optimize operations. On AWS, the company can integrate multiple sources and make organizational data available to shipping companies. To manage this data, it has adopted [Amazon Simple Storage Service](#) (Amazon S3)—an object storage service built to retrieve any amount of data from anywhere—and [Amazon Relational Database Service](#) (Amazon RDS), a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud. On Amazon S3, data is securely stored and readily accessible, while Amazon RDS helps organize this information so that shipping companies can extract actionable insights.

With serverless technologies such as [AWS Fargate](#), a serverless, pay-as-you-go compute engine, and [AWS Lambda](#), an event-driven compute service, ZeroNorth can run its software without having to manage every component of its infrastructure. These solutions, combined with containers running on [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed Kubernetes service, empower the company to scale its digital services efficiently, even as demand grows. In fact, ZeroNorth has introduced five new services in one year and increased the adoption of its solution from a few hundred vessels to over 4,000.

On AWS, ZeroNorth developers can rapidly test and iterate new solutions, accelerating the company's speed of innovation. The cloud-based infrastructure makes it simple for the company to explore new tools, fail fast, and learn quickly using machine learning and AWS analytics. As a result, ZeroNorth can scale its business and get new services out to market at a much faster pace.

For ZeroNorth, these benefits go far beyond business growth. By using AWS to digitize fleet operations, the company is providing the shipping industry with better data access and visibility. With these insights, organizations can make decisions to optimize their operations, leading to lower fuel consumption and, in turn, reduced emissions. ZeroNorth's approach makes it possible for stakeholders to engage in effective decarbonization initiatives and meet the urgent call to action posed by climate change.

Outcome | Advancing the Shipping Industry toward a More Sustainable Future

On AWS, ZeroNorth is helping the shipping industry progress toward a greener, cleaner future. With its data and services now in the cloud, the company is exploring new business areas to foster a positive environmental impact. Looking ahead, ZeroNorth plans to incorporate high-frequency, near-real-time data from vessels to augment the breadth of information available on its software. This expansion will help shipping companies optimize even further, making sure that their vessels operate at peak efficiency.

"Our ambition is to support the entire shipping industry with their optimization needs to reach ambitious and necessary decarbonization targets," says Martin Seest Christiansen, vice president of platform engineering at ZeroNorth. "Huge ambitions require capable technology, and that's what we have in AWS."

About ZeroNorth

Headquartered in Copenhagen, Denmark, ZeroNorth helps shipping companies optimize their centralized digital solutions. Over 4,000 vessels rely on ZeroNorth's technology to work more sustainably.



AWS Services Used

Amazon Simple Storage Service (Amazon S3)

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon Relational Database Service (Amazon RDS)

Amazon RDS is a collection of managed services that makes it simple to set up, operate, and scale databases in the cloud.

[Learn more »](#)

Amazon Elastic Kubernetes Service (Amazon EKS)

Amazon EKS is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

[Learn more »](#)

AWS Fargate

AWS Fargate is a serverless, pay-as-you-go compute engine that lets you focus on building applications without managing servers.

[Learn more »](#)



HHLA Sky Uses AWS to Scale and Secure IoT...

HHLA Sky GmbH, an autonomous drone and mobile robot developer, uses Amazon Web Services (AWS) to scale its solutions to customers globally. The ability to run its Internet of Things (IoT) software in secure environments is vital, says CEO Matthias Gronstedt. Hear how



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Driving Business with Advanced Monitoring and...

After modernizing its solutions on AWS, container shipping line Hapag-Lloyd AG built a central logging system within its Web & Mobile team to improve observability, facilitating better business decisions.

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Portbase Keeps Digital Port Logistics Running...

Portbase migrated its data platform to AWS so its developers could have complete control over its products and manage its services end-to-end. During global supply chain disruption, its platform helps keep goods on the



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IBS Software Leverages AWS Enterprise Support...

IBS Software Leverages AWS Enterprise Support to Deliver Industry-Transforming Products for the Travel and Hospitality Industry

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